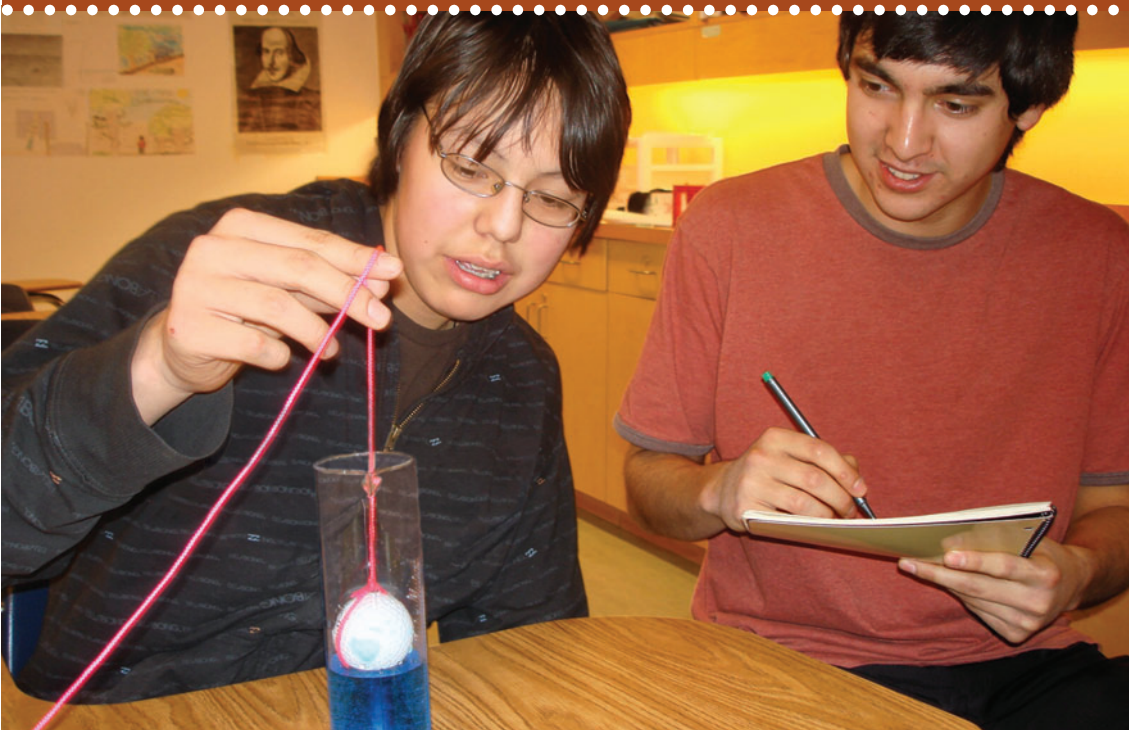


4.6 Applying the Exponent Laws



LESSON FOCUS

Apply the exponent laws to simplify expressions.

We can use a measuring cylinder to determine the volume of a sphere. Then we can use the exponent laws to help calculate the radius.

Make Connections

Recall the exponent laws for integer bases and whole number exponents.

Product of powers: $a^m \cdot a^n = a^{m+n}$

Quotient of powers: $a^m \div a^n = a^{m-n}$, $a \neq 0$

Power of a power: $(a^m)^n = a^{mn}$

Power of a product: $(ab)^m = a^m b^m$

Power of a quotient: $\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}$, $b \neq 0$

What other types of numbers could be a base? An exponent?

How would you use the exponent laws to evaluate an expression with these numbers?

Construct Understanding

THINK ABOUT IT

Work on your own.

What is the value of $\left(\frac{a^6 b^9}{a^5 b^8}\right)^{-2}$ when $a = -3$ and $b = 2$?

Compare strategies with a classmate.

If you used the same strategy, find a different strategy.

Which strategy is more efficient, and why?

We can use the exponent laws to simplify expressions that contain rational number bases. It is a convention to write a simplified power with a positive exponent.

Example 1 Simplifying Numerical Expressions with Rational Number Bases

Simplify by writing as a single power. Explain the reasoning.

$$\begin{array}{ll} \text{a) } 0.3^{-3} \cdot 0.3^5 & \text{b) } \left[\left(-\frac{3}{2} \right)^{-4} \right]^2 \cdot \left[\left(-\frac{3}{2} \right)^2 \right]^3 \\ \text{c) } \frac{(1.4^3)(1.4^4)}{1.4^{-2}} & \text{d) } \left(\frac{7^{\frac{2}{3}}}{7^{\frac{1}{3}} \cdot 7^{\frac{5}{3}}} \right)^6 \end{array}$$

SOLUTION

$$\text{a) } 0.3^{-3} \cdot 0.3^5$$

Use the product of powers law:

When the bases are the same, add the exponents.

$$\begin{aligned} 0.3^{-3} \cdot 0.3^5 &= 0.3^{(-3) + 5} \\ &= 0.3^2 \end{aligned}$$

$$\text{b) } \left[\left(-\frac{3}{2} \right)^{-4} \right]^2 \cdot \left[\left(-\frac{3}{2} \right)^2 \right]^3$$

First use the power of a power law:

For each power, multiply the exponents.

$$\left[\left(-\frac{3}{2} \right)^{-4} \right]^2 \cdot \left[\left(-\frac{3}{2} \right)^2 \right]^3 = \left(-\frac{3}{2} \right)^{(-4)(2)} \cdot \left(-\frac{3}{2} \right)^{(2)(3)}$$

Then use the product of powers law.

$$\begin{aligned} \left[\left(-\frac{3}{2} \right)^{-4} \right]^2 \cdot \left[\left(-\frac{3}{2} \right)^2 \right]^3 &= \left(-\frac{3}{2} \right)^{-8} \cdot \left(-\frac{3}{2} \right)^6 \\ &= \left(-\frac{3}{2} \right)^{-2} && \text{Write with a positive exponent.} \\ &= \left(-\frac{2}{3} \right)^2 \end{aligned}$$

$$\text{c) } \frac{(1.4^3)(1.4^4)}{1.4^{-2}}$$

Use the product of powers law.

$$= \frac{1.4^{3+4}}{1.4^{-2}}$$

$$= \frac{1.4^7}{1.4^{-2}}$$

Use the quotient of powers law.

$$= 1.4^{7 - (-2)}$$

$$= 1.4^9$$

(Solution continues.)

CHECK YOUR UNDERSTANDING

1. Simplify by writing as a single power. Explain your reasoning.

$$\text{a) } 0.8^2 \cdot 0.8^{-7}$$

$$\text{b) } \left[\left(-\frac{4}{5} \right)^2 \right]^{-3} \div \left[\left(-\frac{4}{5} \right)^4 \right]^{-5}$$

$$\text{c) } \frac{(1.5^{-3})^{-5}}{1.5^5}$$

$$\text{d) } \frac{9^{\frac{5}{4}} \cdot 9^{-\frac{1}{4}}}{9^{\frac{3}{4}}}$$

$$[\text{Answers: a) } \frac{1}{0.8^5} \quad \text{b) } \left(-\frac{4}{5} \right)^{14}]$$

$$\text{c) } 1.5^{10} \quad \text{d) } 9^{\frac{1}{4}}$$

$$\begin{aligned}
 \text{d) } & \left(\frac{7^{\frac{2}{3}}}{7^{\frac{1}{3}} \cdot 7^{\frac{5}{3}}} \right)^6 && \text{Use the product of powers law.} \\
 & = \left(\frac{7^{\frac{2}{3}}}{7^{\frac{1}{3} + \frac{5}{3}}} \right)^6 \\
 & = \left(\frac{7^{\frac{2}{3}}}{7^{\frac{6}{3}}} \right)^6 && \text{Use the quotient of powers law.} \\
 & = \left(7^{\frac{2}{3} - \frac{6}{3}} \right)^6 \\
 & = \left(7^{-\frac{4}{3}} \right)^6 && \text{Use the power of a power law.} \\
 & = 7^{\left(-\frac{4}{3}\right)(6)} \\
 & = 7^{-\frac{24}{3}} \\
 & = 7^{-8} && \text{Write with a positive exponent.} \\
 & = \frac{1}{7^8}
 \end{aligned}$$

In part d, what other strategy could you use? Which strategy is more efficient?

Example 2 Simplifying Algebraic Expressions with Integer Exponents

Simplify. Explain the reasoning.

$$\text{a) } (x^3y^2)(x^2y^{-4}) \qquad \text{b) } \frac{10a^5b^3}{2a^2b^{-2}}$$

SOLUTION

$$\begin{aligned}
 \text{a) } (x^3y^2)(x^2y^{-4}) &= x^3 \cdot y^2 \cdot x^2 \cdot y^{-4} && x^3y^2 \text{ means } x^3 \cdot y^2 \\
 &= x^3 \cdot x^2 \cdot y^2 \cdot y^{-4} && \text{Use the product of} \\
 &= x^{3+2} \cdot y^{2+(-4)} && \text{powers law.} \\
 &= x^5 \cdot y^{-2} && \text{Write with a positive exponent.} \\
 &= x^5 \cdot \frac{1}{y^2} \\
 &= \frac{x^5}{y^2}
 \end{aligned}$$

$$\begin{aligned}
 \text{b) } \frac{10a^5b^3}{2a^2b^{-2}} &= \frac{10}{2} \cdot \frac{a^5}{a^2} \cdot \frac{b^3}{b^{-2}} && \text{Use the quotient of powers law.} \\
 &= 5 \cdot a^{5-2} \cdot b^{3-(-2)} \\
 &= 5 \cdot a^3 \cdot b^5 \\
 &= 5a^3b^5
 \end{aligned}$$

CHECK YOUR UNDERSTANDING

2. Simplify. Explain your reasoning.

$$\text{a) } m^4n^{-2} \cdot m^2n^3$$

$$\text{b) } \frac{6x^4y^{-3}}{14xy^2}$$

$$[\text{Answers: a) } m^6n \quad \text{b) } \frac{3x^3}{7y^5}]$$

Example 3**Simplifying Algebraic Expressions with Rational Exponents**

Simplify. Explain the reasoning.

a) $(8a^3b^6)^{\frac{1}{3}}$

b) $(x^{\frac{3}{2}}y^2)(x^{\frac{1}{2}}y^{-1})$

c) $\frac{4a^{-2}b^{\frac{2}{3}}}{2a^2b^{\frac{1}{3}}}$

d) $\left(\frac{100a}{25a^5b^{-\frac{1}{2}}}\right)^{\frac{1}{2}}$

SOLUTION

a) $(8a^3b^6)^{\frac{1}{3}} = 8^{\frac{1}{3}} \cdot a^{3(\frac{1}{3})} \cdot b^{6(\frac{1}{3})}$

Using the power of a power law.

$$= (2^3)^{\frac{1}{3}} \cdot a^1 \cdot b^2$$

$$= 2ab^2$$

b) $(x^{\frac{3}{2}}y^2)(x^{\frac{1}{2}}y^{-1}) = x^{\frac{3}{2}} \cdot x^{\frac{1}{2}} \cdot y^2 \cdot y^{-1}$

Use the product of powers law.

$$= x^{\frac{3}{2} + \frac{1}{2}} \cdot y^{2 + (-1)}$$

$$= x^2y$$

c) $\frac{4a^{-2}b^{\frac{2}{3}}}{2a^2b^{\frac{1}{3}}} = \frac{4}{2} \cdot \frac{a^{-2}}{a^2} \cdot \frac{b^{\frac{2}{3}}}{b^{\frac{1}{3}}}$

Use the quotient of powers law.

$$= 2 \cdot a^{(-2) - 2} \cdot b^{\frac{2}{3} - \frac{1}{3}}$$

$$= 2 \cdot a^{-4} \cdot b^{\frac{1}{3}}$$

Write with a positive exponent.

$$= \frac{2b^{\frac{1}{3}}}{a^4}$$

d) $\left(\frac{100a}{25a^5b^{-\frac{1}{2}}}\right)^{\frac{1}{2}} = \left(\frac{100}{25} \cdot \frac{a^1}{a^5} \cdot \frac{1}{b^{-\frac{1}{2}}}\right)^{\frac{1}{2}}$

Simplify inside the brackets first.

Use the quotient of powers law.

Write with a positive exponent.

$$= \left(4 \cdot a^{1-5} \cdot b^{\frac{1}{2}}\right)^{\frac{1}{2}}$$

$$= \left(4 \cdot a^{-4} \cdot b^{\frac{1}{2}}\right)^{\frac{1}{2}}$$

Use the power of a power law.

$$= 4^{\frac{1}{2}} \cdot a^{(-4)(\frac{1}{2})} \cdot b^{(\frac{1}{2})(\frac{1}{2})}$$

$$= 2 \cdot a^{-2} \cdot b^{\frac{1}{4}}$$

Write with a positive exponent.

$$= \frac{2b^{\frac{1}{4}}}{a^2}$$

CHECK YOUR UNDERSTANDING

3. Simplify. Explain your reasoning.

a) $(25a^4b^2)^{\frac{3}{2}}$

b) $(x^3y^{-\frac{3}{2}})(x^{-1}y^{\frac{1}{2}})$

c) $\frac{12x^{-5}y^{\frac{5}{2}}}{3x^{\frac{1}{2}}y^{-\frac{1}{2}}}$

d) $\left(\frac{50x^2y^4}{2x^4y^7}\right)^{\frac{1}{2}}$

[Answers: a) $125a^6b^3$ b) $\frac{x^2}{y}$

c) $\frac{4y^3}{x^{\frac{11}{2}}}$ d) $\frac{5}{xy^{\frac{3}{2}}}$

Example 4

Solving Problems Using the Exponent Laws

A sphere has volume 425 m^3 .

What is the radius of the sphere to the nearest tenth of a metre?

SOLUTION

The volume V of a sphere with radius r is given by the formula: $V = \frac{4}{3}\pi r^3$. Substitute $V = 425$, then solve for r .

$$425 = \frac{4}{3}\pi r^3 \quad \text{Multiply each side by 3.}$$

$$3(425) = 3\left(\frac{4}{3}\pi r^3\right)$$

$$1275 = 4\pi r^3 \quad \text{Divide each side by } 4\pi.$$

$$\frac{1275}{4\pi} = \frac{4\pi r^3}{4\pi}$$

$$\frac{1275}{4\pi} = r^3 \quad \text{To solve for } r, \text{ take the cube root of each side by raising each side to the one-third power.}$$

$$\left(\frac{1275}{4\pi}\right)^{\frac{1}{3}} = (r^3)^{\frac{1}{3}} \quad \text{Use the power of a power law.}$$

$$\left(\frac{1275}{4\pi}\right)^{\frac{1}{3}} = r$$
$$r = 4.6640\dots$$

The radius of the sphere is approximately 4.7 m.

CHECK YOUR UNDERSTANDING

4. A cone with height and radius equal has volume 18 cm^3 . What are the radius and height of the cone to the nearest tenth of a centimetre?

[Answer: approximately 2.6 cm]

How do you know that the length of the radius is an irrational number?

Discuss the Ideas

1. Suppose you want to evaluate an algebraic expression for particular values of the variables. Why might it be helpful to simplify the expression first?
2. When you simplify an expression, how do you know which exponent law to apply first?

Exercises

Write all powers with positive exponents.

A

3. Simplify.

a) $x^3 \cdot x^4$ b) $a^2 \cdot a^{-5}$
c) $b^{-3} \cdot b^5$ d) $m^2 \cdot m^{-3}$

4. Write as a single power.

a) $0.5^2 \cdot 0.5^3$ b) $0.5^2 \cdot 0.5^{-3}$
c) $\frac{0.5^2}{0.5^3}$ d) $\frac{0.5^2}{0.5^{-3}}$

5. Simplify.

a) $\frac{x^4}{x^2}$ b) $\frac{x^2}{x^5}$
c) $n^6 \div n^5$ d) $\frac{a^2}{a^6}$

6. Simplify.

a) $(n^2)^3$ b) $(z^2)^{-3}$
c) $(n^{-4})^{-3}$ d) $(c^{-2})^2$

7. Write as a single power.

$$\begin{array}{ll} \text{a) } \left[\left(\frac{3}{5} \right)^3 \right]^4 & \text{b) } \left[\left(\frac{3}{5} \right)^3 \right]^{-4} \\ \text{c) } \left[\left(\frac{3}{5} \right)^{-3} \right]^{-4} & \text{d) } \left[\left(-\frac{3}{5} \right)^{-3} \right]^{-4} \end{array}$$

8. Simplify.

$$\begin{array}{ll} \text{a) } \left(\frac{a}{b} \right)^2 & \text{b) } \left(\frac{n^2}{m} \right)^3 \\ \text{c) } \left(\frac{c^2}{d^2} \right)^{-4} & \text{d) } \left(\frac{2b}{5c} \right)^2 \\ \text{e) } (ab)^2 & \text{f) } (n^2m)^3 \\ \text{g) } (c^3d^2)^{-4} & \text{h) } (xy^{-1})^3 \end{array}$$

B

9. Simplify. State the exponent law you used.

$$\begin{array}{ll} \text{a) } x^{-3} \cdot x^4 & \text{b) } a^{-4} \cdot a^{-1} \\ \text{c) } b^4 \cdot b^{-3} \cdot b^2 & \text{d) } m^8 \cdot m^{-2} \cdot m^{-6} \\ \text{e) } \frac{x^{-5}}{x^2} & \text{f) } \frac{s^5}{s^{-5}} \\ \text{g) } \frac{b^{-8}}{b^{-3}} & \text{h) } \frac{t^{-4}}{t^{-4}} \end{array}$$

10. Evaluate.

$$\begin{array}{ll} \text{a) } 1.5^{\frac{3}{2}} \cdot 1.5^{\frac{1}{2}} & \text{b) } \left(\frac{3}{4} \right)^{\frac{3}{4}} \cdot \left(\frac{3}{4} \right)^{\frac{5}{4}} \\ \text{c) } (-0.6)^{\frac{1}{3}} \cdot (-0.6)^{\frac{5}{3}} & \text{d) } \left(\frac{4}{5} \right)^{\frac{4}{3}} \cdot \left(\frac{4}{5} \right)^{\frac{4}{3}} \\ \text{e) } \frac{0.6^{\frac{1}{2}}}{0.6^{\frac{3}{2}}} & \text{f) } \frac{\left(-\frac{3}{8} \right)^{\frac{2}{3}}}{\left(-\frac{3}{8} \right)^{\frac{1}{3}}} \\ \text{g) } \frac{0.49^{\frac{5}{2}}}{0.49^4} & \text{h) } \frac{0.027^{\frac{5}{3}}}{0.027^{\frac{4}{3}}} \end{array}$$

11. Simplify. Explain your reasoning.

$$\begin{array}{ll} \text{a) } (x^{-1}y^{-2})^{-3} & \text{b) } (2a^{-2}b^2)^{-2} \\ \text{c) } (4m^2n^3)^{-3} & \text{d) } \left(\frac{3}{2}m^{-2}n^{-3} \right)^{-4} \end{array}$$

12. A cone with equal height and radius has volume 1234 cm^3 . What is the height of the cone to the nearest tenth of a centimetre?

13. A sphere has volume 375 cubic feet. What is the surface area of the sphere to the nearest square foot?

14. Simplify. Which exponent laws did you use?

$$\text{a) } \frac{(a^2b^{-1})^{-2}}{(a^{-3}b)^3} \quad \text{b) } \frac{(c^{-3}d)^{-1}}{c^2d}^{-2}$$

15. Evaluate each expression for $a = -2$ and $b = 1$. Explain your strategy.

$$\begin{array}{ll} \text{a) } (a^3b^2)(a^2b^3) & \text{b) } (a^{-1}b^{-2})(a^{-2}b^{-3}) \\ \text{c) } \frac{a^{-4}b^5}{ab^3} & \text{d) } \left(\frac{a^{-7}b^7}{a^{-9}b^{10}} \right)^{-5} \end{array}$$

16. Simplify.

$$\begin{array}{ll} \text{a) } m^{\frac{2}{3}} \cdot m^{\frac{4}{3}} & \text{b) } x^{\frac{3}{2}} \div x^{\frac{1}{4}} \\ \text{c) } \frac{-9a^{-4}b^{\frac{3}{4}}}{3a^2b^{\frac{1}{4}}} & \text{d) } \left(\frac{-64c^6}{a^9b^{-\frac{1}{2}}} \right)^{\frac{1}{3}} \end{array}$$

17. Identify any errors in each solution for simplifying an expression. Write a correct solution.

$$\begin{array}{l} \text{a) } (x^2y^{-3})(x^{\frac{1}{2}}y^{-1}) = x^2 \cdot x^{\frac{1}{2}} \cdot y^{-3} \cdot y^{-1} \\ \quad = x^{\frac{5}{2}} \cdot y^{-4} \\ \quad = xy^3 \\ \text{b) } \left(\frac{-5a^2}{b^{\frac{1}{2}}} \right)^{-2} = \frac{10a^{-4}}{b^{-1}} \\ \quad = \frac{10b}{a^4} \end{array}$$

18. Explain how to use a measuring cylinder containing water to calculate the diameter of a marble that fits inside the cylinder.

19. Identify the errors in each simplification. Write the correct solution.

$$\begin{array}{l} \text{a) } \frac{(m^{-3} \cdot n^2)^{-4}}{(m^2 \cdot n^{-3})^2} = (m^{-5} \cdot n^5)^{-6} \\ \quad = m^{30} \cdot n^{30} \\ \quad = (mn)^{30} \\ \text{b) } \left(\frac{1}{r^2} \cdot s^{-\frac{3}{2}} \right)^{\frac{1}{2}} \cdot \left(r^{-\frac{1}{4}} \cdot s^{\frac{1}{2}} \right)^{-1} = r^1 \cdot s^{-1} \cdot r^{-\frac{5}{4}} \cdot s^{-\frac{1}{2}} \\ \quad = r^{1-\frac{5}{4}} \cdot s^{-1-\frac{1}{2}} \\ \quad = r^{-\frac{1}{4}} \cdot s^{-\frac{3}{2}} \\ \quad = \frac{1}{r^{\frac{1}{4}} \cdot s^{\frac{3}{2}}} \end{array}$$

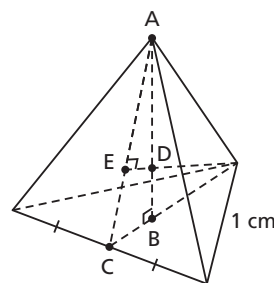
20. ISO paper sizes A0, A1, A2, ..., are commonly used outside of North America. For any whole number n , the width, in metres, of a piece of A n paper is $2^{\frac{2n+1}{4}}$ and its length, in metres, is $2^{\frac{2n-1}{4}}$.
- Write, then simplify expressions to represent the dimensions of each piece of paper. Evaluate each measure to the nearest millimetre.
 - A3
 - A4
 - A5
 - Suppose each piece of paper from part a is folded in half along a line perpendicular to its length. Write, then simplify expressions to represent the dimensions of each folded piece.
 - Compare your results in parts a and b. What do you notice?

C

21. Simplify. Show your work.
- $\left(\frac{a^{-3}b}{c^2}\right)^{-4} \cdot \left(\frac{c^5}{a^4b^{-3}}\right)^{-1}$
 - $\frac{(2a^{-1}b^4c^{-3})^{-2}}{(4a^2bc^{-4})^2}$
22. If $x = a^{-2}$ and $y = a^{\frac{2}{3}}$, write each expression in terms of a .
- $\left(x^{\frac{1}{2}}y^{\frac{2}{3}}\right)^2$
 - $\left(x^{\frac{3}{4}} \div y^{-\frac{1}{2}}\right)^3$

23. Write 3 different expressions for each result.
- $x^{\frac{3}{2}}$ is the product of two powers with rational exponents.
 - $x^{\frac{3}{2}}$ is the quotient of two powers with rational exponents.
 - $x^{\frac{3}{2}}$ is the result of raising a power with a rational exponent to a rational exponent.
24. A regular tetrahedron has edge length 1 cm. It is placed inside a sphere so that all its vertices touch the surface of the sphere. Point D is the centre of the sphere. The measures, in centimetres, of 3 line segments are:

$$AB = \left(\frac{2}{3}\right)^{\frac{1}{2}}; AC = \frac{3}{2}\left(\frac{1}{3}\right)^{\frac{1}{2}}; AE = \left(\frac{1}{3}\right)^{\frac{1}{2}}$$



Given that $\triangle ABC$ is similar to $\triangle AED$ and $\frac{AC}{AB} = \frac{AD}{AE}$, determine the length of AD.

Reflect

Explain how to apply the exponent laws to simplify algebraic expressions. Use examples to illustrate the types of expressions you can simplify.



THE WORLD OF MATH

Math Fact: Platonic Solids

Platonic solids are the only regular polyhedra that can be placed in a sphere so that each vertex touches the surface of the sphere. In about 300 B.C.E., Euclid used trigonometry, similar triangles, and the Pythagorean Theorem to show the ratio of the edge length of each Platonic solid to the diameter of the sphere:

Tetrahedron



$$1 : \left(\frac{3}{2}\right)^{\frac{1}{2}}$$

Cube



$$1 : 3^{\frac{1}{2}}$$

Octahedron



$$1 : 2^{\frac{1}{2}}$$

Dodecahedron



$$1 : \frac{1}{2} \left(3^{\frac{1}{2}} + 15^{\frac{1}{2}} \right)$$

Icosahedron



$$1 : \frac{1}{2} \left(10 + 2(5)^{\frac{1}{2}} \right)^{\frac{1}{2}}$$